

Three LES cases

BOMEX (ocean), SCMS and ARM (land).

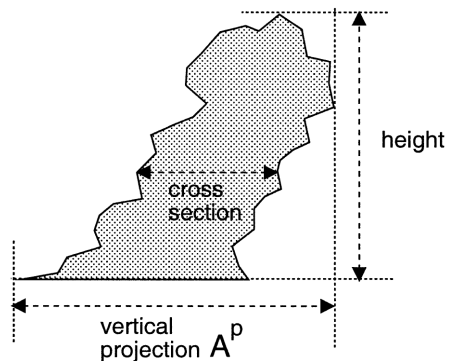
Cloud: connected saturated cells.

$$l = \sqrt{A^p}$$

$$N = \int_0^\infty \mathcal{N}(l) dl$$

$\mathcal{N}(l)$: cloud count per unit size.

A^p : vertically projected cloud area.



Cloud object



Projection A^p

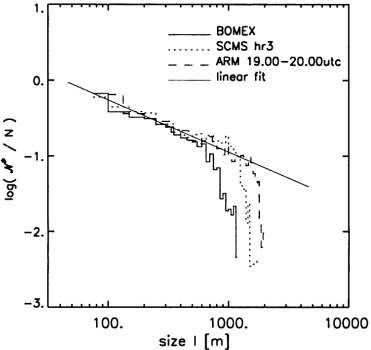


Size histogram

Neggers et al. (2003): cloud-size statistics

Size Statistics of Cumulus Cloud Populations in Large-Eddy Simulations, Neggers, Jonker and Siebesma (2003). The study measures the sizes of shallow cumulus clouds in LES and their contributions to cloud cover and vertical transport. BOMEX represents steady trade-wind cumulus over the ocean. SCMS and ARM represent continental shallow cumulus, with a diurnal cycle in ARM. The reference horizontal grid spacing is 50 m, in a 6.4 km domain. Repeated snapshots sample tens of thousands of clouds per case.

A cloud is a connected set of saturated grid cells. Figure 3 shows the distinction between a cross-section at one height and the vertical projection of the whole cloud. The chosen size is $l = \sqrt{A^p}$, not a radius or a diameter. The size density $\mathcal{N}(l)$ gives counts per unit size, and its integral is the total count. The scheme summarizes the actual measurement procedure. Source: Sections 2–3 and Appendix, Figure 3, Equations (1)–(2).



Size density below ℓ_d

$$\mathcal{N}(l) \propto l^{-1.70}$$

About one decade.
A steeper fall above ℓ_d .

Break ℓ_d	
BOMEX	700 m
SCMS	1050 m
ARM	400–1250 m

$$\mathcal{N}^* = l \ln(10) \mathcal{N} \quad (\text{log bins})$$

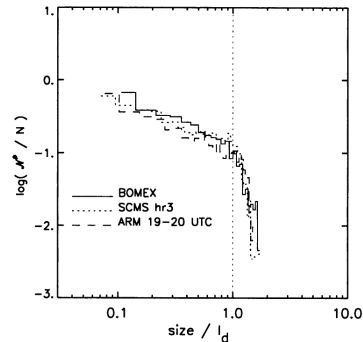
The break varies with case and time of day

A power law below a variable size break

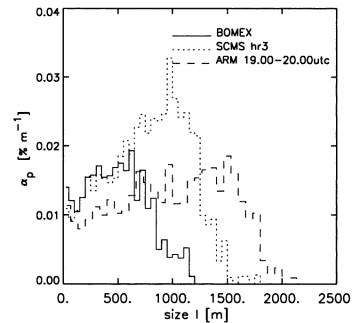
Figure 4a shows the normalized size density in logarithmic bins. Below the scale break, the three cases have approximately the same slope. The fitted log-bin slope is -0.70 , corresponding to the exponent $b = -1.70$ of $\mathcal{N}(l) \propto l^b$ because $\mathcal{N}^* = l \ln(10) \mathcal{N}$. The supported range spans roughly one decade between the grid scale and the break.

Above ℓ_d , the density decreases faster. Table 2 gives 700 m for BOMEX and 1050 m for SCMS. During the ARM afternoon, the break increases from 400 to 1250 m. Increasing the number of sampled clouds does not remove the break. Its physical origin remains unresolved. It is a cloud-size scale, not an inter-cloud distance. Source: Section 4a, Figure 4a, Table 2, Equations (8)–(9).

Rescaling by ℓ_d : collapse



$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{S}$$



Area weighting gives $\alpha^p \propto l^{b+2}$: the cover peaks near ℓ_d

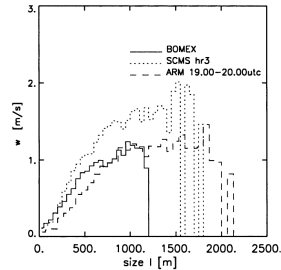
The size break sets the dominant cloud cover

Left: Figure 6 plots the three size densities against l/ℓ_d . Their approximate collapse supports a common functional form for these simulated cumulus populations, with a case-dependent break. The scaling range and three tested cases limit this claim.

Right: Figure 7a gives the contribution to projected cloud cover per unit size. Each cloud contributes its projected area l^2 , so $\alpha^p(l) = l^2 \mathcal{N}(l)/S$, where S is the horizontal domain area. For a power law with exponent b , the cover decomposition scales as l^{b+2} . Below the break, $b = -1.70$ gives a positive exponent, 0.30. Above the break the density declines sufficiently rapidly that the cover decomposition decreases. The peak therefore lies at an intermediate size near the break.

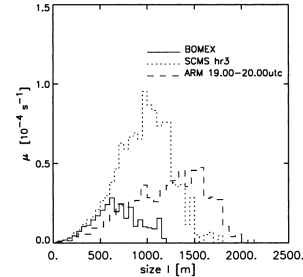
This is a decomposition per unit linear size, not per logarithmic bin. The total projected cover is $a^p = \int_0^\infty \alpha^p(l) dl$. Source: Section 3, Section 4a-b, Figures 6 and 7a, Equations (3)–(4) and (10).

Mean vertical velocity



Small clouds carry little flux.

Mass-flux contribution



$$\mu(l) = \overline{\alpha(l, z) w(l, z)}$$

Mass flux peaks at larger cloud sizes than projected cover

Larger cloud velocities shift the transport peak

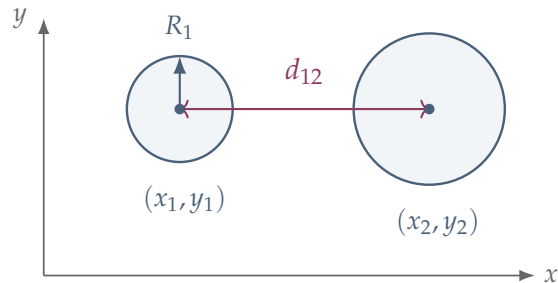
Figure 7b shows that cloud-mean vertical velocity generally increases with cloud size. Figure 7c shows the resulting mass-flux decomposition. At height z , $\mu(l, z) = \alpha(l, z)w(l, z)$, where $\alpha(l, z)$ is the cloud-fraction contribution per unit size. Averaging over the cloud-layer depth h_c gives $\mu(l) = h_c^{-1} \int \alpha(l, z)w(l, z) dz$. Its integral over size is the total kinematic mass flux, in m/s under this paper's convention.

Cloud size is always the projected size, even when evaluating the flux at a specific height. Since w increases with size, the mass-flux maximum occurs at somewhat larger cloud sizes than the projected-cover maximum. The smallest clouds contribute very little to total vertical transport.

The intermediate dominant size persists for tested grid spacings of 25–100 m. A 3.2 km domain truncates the largest clouds. Increased shear broadens projected sizes while leaving total mass flux almost unchanged. These sensitivity tests support the main transport result but do not identify the mechanism of the size break. Source: Section 3, Section 4b–e, Figure 7b–c and Figures 9–11.

Reported: projected cloud sizes, $\mathcal{N}(l)$.

Spatial quantities (schematic)



Not reported for thermals

Size distribution R_i

Positions (x_i, y_i)

Distances d_{ij}

Projected cumulus size does not determine thermal size or spacing

Scope: §3–5 and Appendix. The sketch defines the quantities sought. It represents no observed or simulated thermal field.

No 2D distribution of thermals

Our reading sought distributions of thermals in a horizontal plane: their size at a fixed height, their positions and their separations. The diagram only defines these three quantities. Its circles, radii and separation are schematic, not data or a spatial model.

Neggers et al. report projected cloud sizes and cloud transport statistics. A cloud is a saturated object, whereas a thermal can be unsaturated and can feed a cloud without sharing its projected footprint. Projection also combines geometry over height and responds to tilt and shear. The paper does not establish a cloud-to-thermal size mapping and does not report thermal centres or inter-thermal distances. The exponent -1.70 and the break ℓ_d therefore remain results about the measured cumulus population. Source: Sections 3–5 and Appendix.